
Design Integration Solutions

Given a use case, identify the trade-offs, limitations, and constraints that meet the proposed solution.

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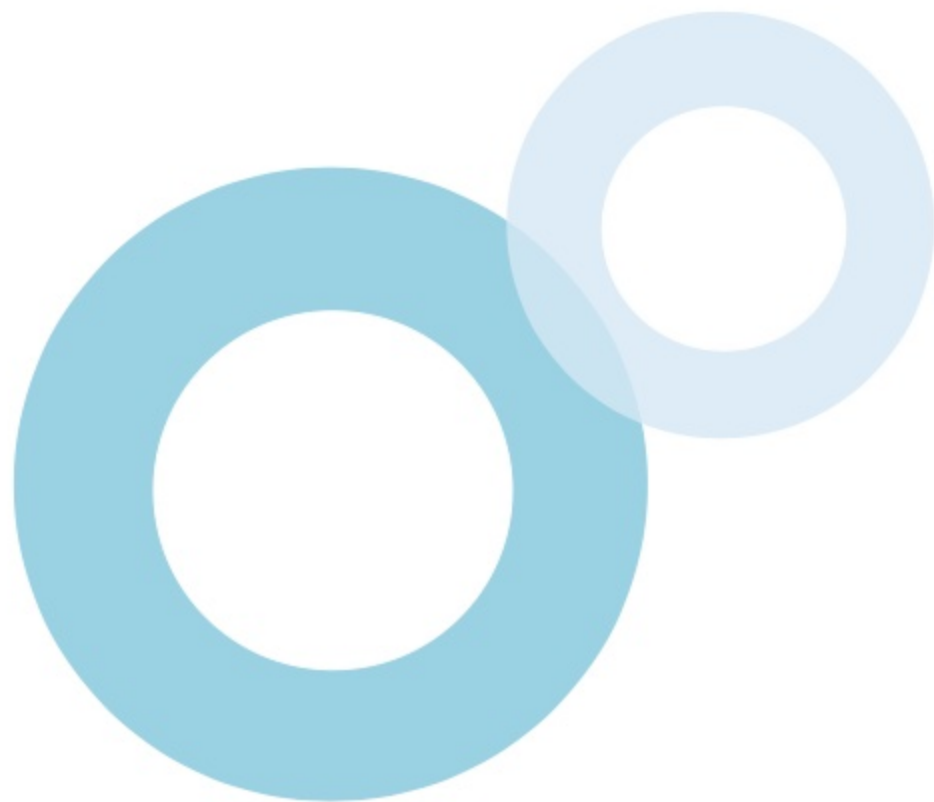


[Requirements & Solutions](#)



After studying this topic, you should be able to:

-  Define a trade-off in an integration solution design and identify the factors that influence it.
-  Identify how limitations and constraints can influence an integration solution.
-  Identify the trade-offs, limitations, and constraints in the given business requirements.



Introduction

An integration solution may be affected by **limitations** or **constraints**. A **trade-off** may be necessary in an integration solution design to meet a particular requirement. It's a **situational decision** that involves **losing one aspect** of the design in favor of **gaining another aspect**.

Limitations that can affect an integration solution include **governor limits**, such as the maximum size of an HTTP response, and **lack of support** for the required capabilities or standards. **Constraints** can make it necessary to consider an alternative solution. For example, if a company does not have the resources for programmatic development, it may be necessary to choose a declarative solution.



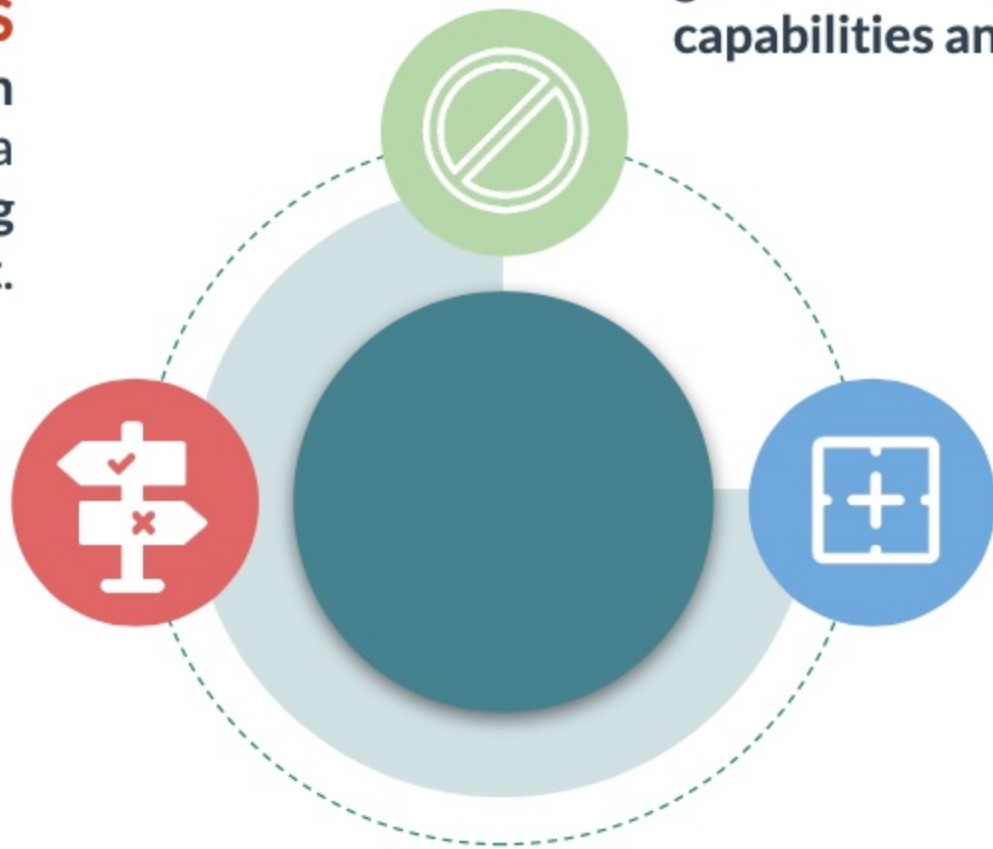
Overview

TRADE-OFFS

A trade-off is a situational decision that involves **losing one aspect** of a solution design in favor of **gaining another aspect**.

LIMITATIONS

An integration solution may be affected by certain **limitations**, such as **governor limits** or **lack of support for capabilities and standards**.



CONSTRAINTS

Constraints, such as **lack of resources for custom development**, can make it necessary to implement an alternative solution.

Integration Trade-Offs

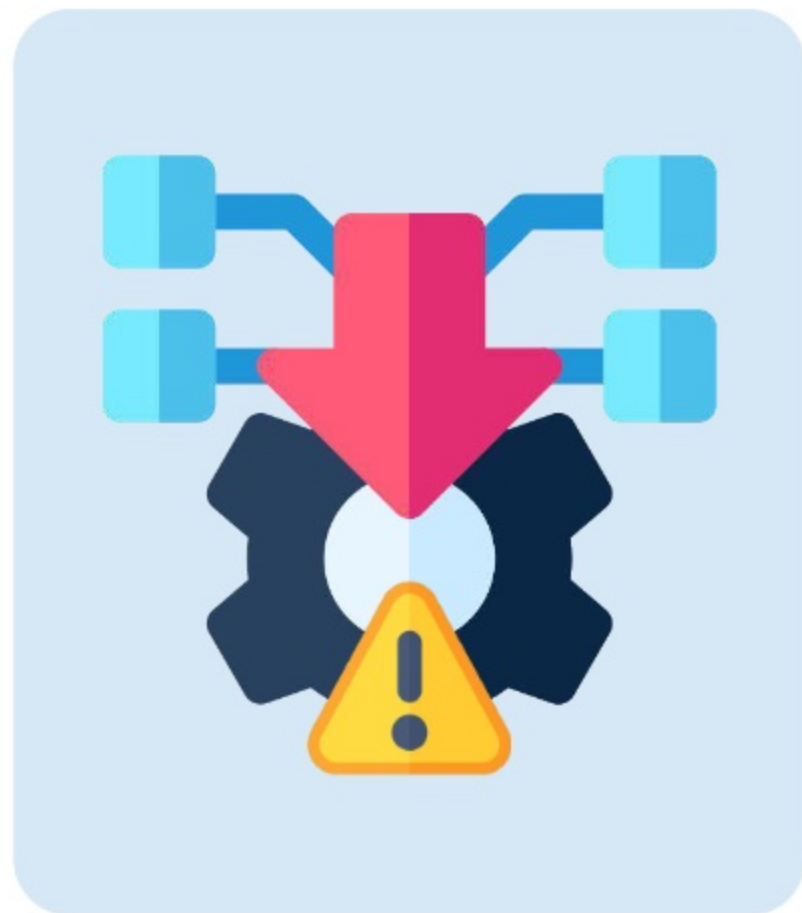
A **trade-off** in an integration solution refers to a **decision** that involves **losing one aspect** of an integration design in return for **gaining another aspect**.

❖ LIMITATION

A **trade-off** generally stems from a **limitation**. For example, if an **Apex HTTP callout** from Salesforce to a remote system takes more than the configurable timeout of up to **120 seconds**, a developer may decide to **reduce the number of steps** performed by the **remote process**.

❖ PROS & CONS

The **pros and cons** of both aspects should be considered before reaching a **decision**. For example, if an integration solution that requires multiple **REST or SOAP API** calls would exceed the **24-hour Salesforce limit**, a developer may decide to **display an error and prevent additional requests** when a user reaches the limit or **postpone the user's request**. The pros and cons of both outcomes should be identified, and the design decision should be made accordingly.



Integration Limitations & Constraints

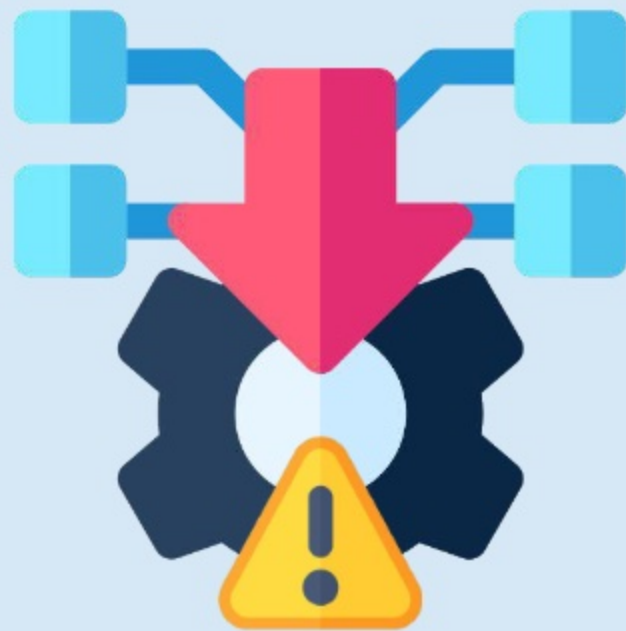
An integration solution may be affected by certain **limitations**. **Constraints** can also affect an integration solution, making it necessary to implement an alternative solution.

❖ SUPPORT

All the **components** of an integration solution must support the required **capabilities and standards**. If there is lack of support, it might be necessary to consider an alternative. For example, if a **remote system is not compatible** with the **web service standards for Apex SOAP callouts**, the use of **Apex HTTP callouts** can be considered.

❖ GOVERNOR LIMITS

An integration solution may be affected by **synchronous or asynchronous governor limits**. For example, **the total number of synchronous or asynchronous HTTP callouts** in a transaction cannot exceed **100**.

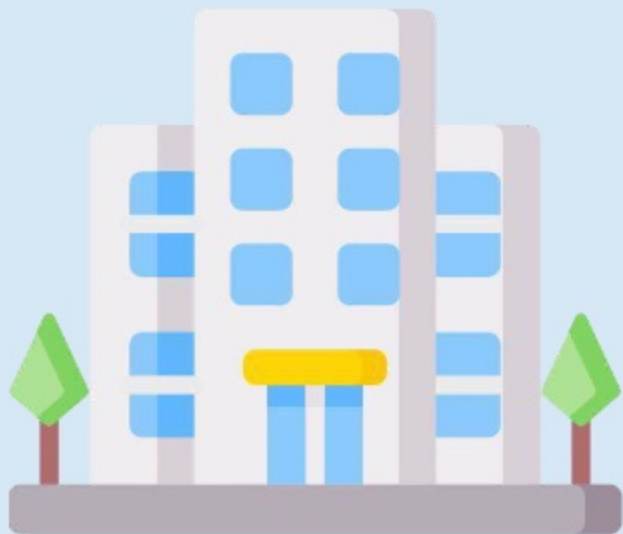


Business Scenario

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Business Scenario: Part 1

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



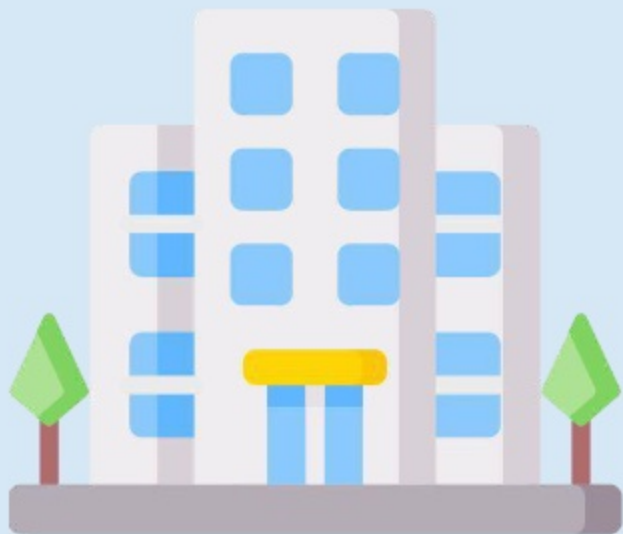
Cosmic Computers is a multinational company that mainly sells products such as laptops, computer components, and printers. Its sales department uses Salesforce primarily to store and manage information about prospects, existing customers, and products.

The internal users of the company also use multiple external systems, including an ERP system, which is the main source of information about the company's resources including products. The support department uses a homegrown customer support management system for case management.

A Workforce Management (WFM) system is used by the HR department to manage employee performance and goals. Furthermore, a Business Intelligence system is used by certain individuals for data analysis.

Business Scenario: Part 2

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Certain developers in the IT department of the company have been working on designing solutions aimed at integrating the systems with one another. However, they have come across various limitations that could affect the integrations once they are implemented.

A solution architect has been hired, whose main task is to identify any trade-offs that could be included in the solution design to circumvent the limitations or constraints.

Requirements & Solutions

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Requirement 1



SCENARIO

A Lightning web component in Salesforce needs to be integrated with the ERP system to allow users in the HR department to retrieve detailed information about a specific employee. The solution design of this integration consists of an Apex HTTP callout to the ERP system, which only supports getting a JSON response containing the data of all the employees. However, the data is likely to exceed the maximum size of the HTTP response, which is 6 MB for synchronous Apex. The solution architect needs to recommend a suitable alternative.



Solution 1



SOLUTION

In this case, the Apex HTTP callout can be executed asynchronously since the maximum size of an HTTP response when using asynchronous Apex is 12 MB. However, this trade-off requires the implementation of a platform event to show the data to the user in near real-time. When the response has been received from the ERP system, a platform event message can be generated. The Lightning web component can subscribe to receive the platform event message containing the data of a specific employee. Once the message has been received, the component can show the employee's information to the current user. Do note, however, that the maximum size of an event message that can be published each time is 1 MB.



Requirement 2



SCENARIO

Data containing any new or updated cases in the support management system should be sent by the support management system to Salesforce using an API call at the end of each day. The current solution design consists of synchronous REST API for creating and updating the necessary data in Salesforce. However, only 200 records can be created or updated in a single API call. But the number of records that require processing could exceed 2,000. The solution architect needs to suggest a suitable option that would allow processing of more than 2,000 records in a single API call.



Solution 2



SOLUTION

Bulk API 2.0 can be used to process more than 2,000 records in a single API call. However, it is important to note that the actual processing of data when using Bulk API 2.0 occurs asynchronously. The existing integration workflow should be modified accordingly. If a user does not need to wait for a response from Salesforce, Bulk API 2.0 should be sufficient.

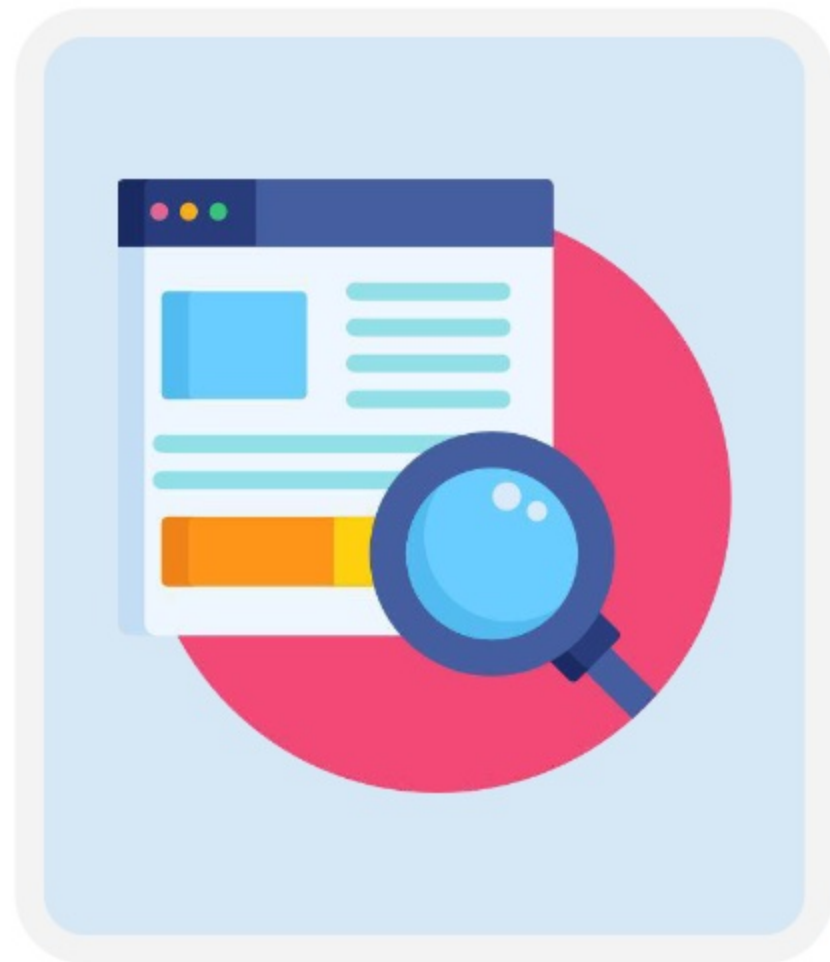


Requirement 3



SCENARIO

The company would like to give certain users from different departments view-only access to performance reviews stored in the Workforce Management (WFM) system. The users who require access are assigned to different profiles. Using Salesforce Connect has been recommended for meeting the requirement. However, since external objects do not support record-level security, the users cannot be given access to external performance review records without making them visible to other users assigned to the same profile. The solution architect needs to recommend a suitable alternative.

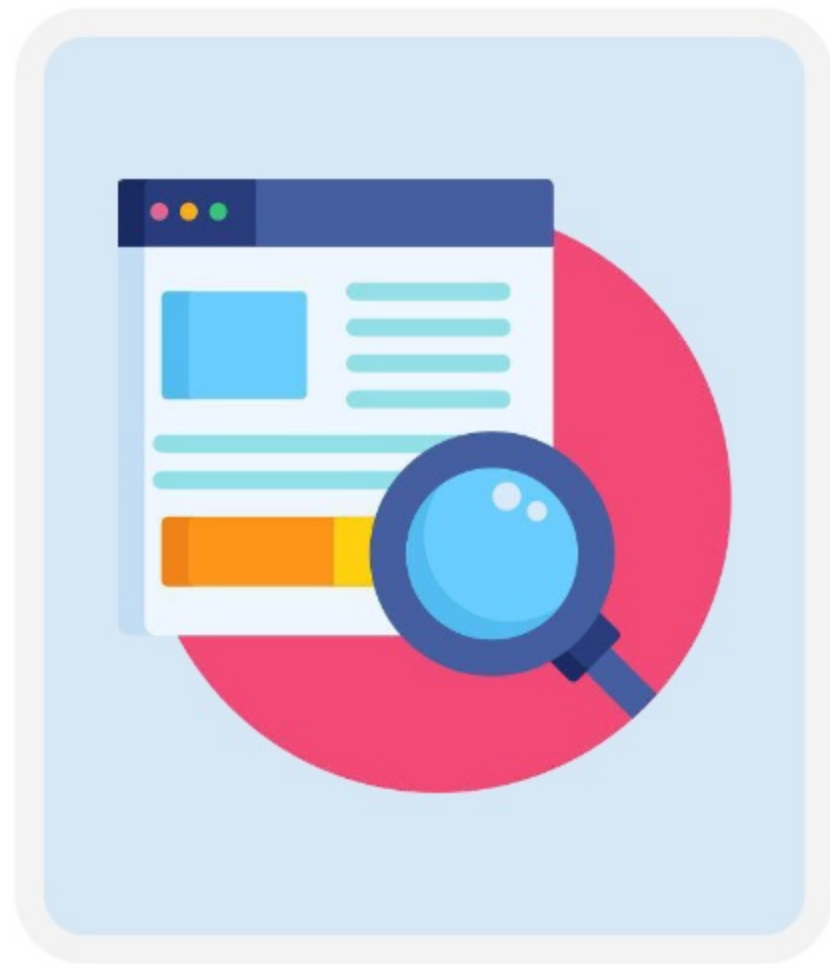


Solution 3



SOLUTION

Instead of configuring Salesforce Connect, an Apex REST or SOAP callout executed in a synchronous manner can be used to retrieve performance reviews from the WFM system. Specific users can be given access to a Lightning web component that executes the callout and allows them to view the external records. It would require programmatic development, but it would limit access to the external data.



Requirement 4



SCENARIO

When an order is created, updated, or deleted in Salesforce, the order information needs to be sent to a Business Intelligence system, which can process SOAP messages, for reporting and analytics. Creating an Apex trigger on the Order object has been recommended for this requirement, but the company does not have the human resources for the programmatic development and testing of the solution. The solution architect needs to recommend an alternative.



Solution 4



SOLUTION

Given the constraint, a declarative approach can be used to meet the requirement. Two flows and an outbound message action can be created. The first flow can be invoked when an order record is created or updated. The second flow can be invoked when an order record is deleted. Both flows can send an outbound message containing order information to the Business Intelligence system.



Learn More



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[Trade-off - Wiki](#)